

In-Class Quiz: April 9

Points: 10/10

✓ **Correct** 1/1 Points

1. What is the first step executed in the microcode preamble? 

- ☐ Decode the instruction
- ☐ Add 4 to the pc
- ☒ Load the instruction register using mloadIR
- ☐ Switch to the first microinstruction using mswitch

✓ **Correct** 1/1 Points

2. Which microinstruction is designed to wait for 1 cycle while the decode registers get populated? 

- ☐ mswitch
- ☒ mdecode
- ☐ mmovi
- ☐ mloadIR

✓ **Correct** 1/1 Points

3. What is the function of the 'mmovi' microinstruction? 

- ☐ It moves data from one register to another register.

- ☒ It moves an immediate value directly into a register.
- ☐ It moves a value from memory into a register.
- ☐ It moves an immediate value to a memory address.

✓ **Correct** 1/1 Points

4. Which of the following best describes the madd microinstruction? 

- ☐ It multiplies two registers together.
- ☐ It adds the values of two source registers.
- ☒ It adds an immediate value to a register.
- ☐ It computes a memory address.

✓ **Correct** 1/1 Points

5. If you need to unconditionally jump to a specific label in your microprogram, which instruction should you use? 

- ☒ mb
- ☐ mbeq
- ☐ mswitch
- ☐ mmov

✓ **Correct** 1/1 Points

6. How is the 'nop' (no operation) instruction implemented in the processor's microcode? 

- ☐ By executing a dummy addition in the ALU.
- ☐ By clearing the contents of all general-purpose registers.

- ☐ By writing a zero to the memory data register.
- ☒ By simply branching back to the preamble using `mb .begin`

✓ **Correct** 1/1 Points

7. In the SimpleRISC microcode architecture, where does the standard 'call' instruction store the return address? 

- ☒ In register 15
- ☐ In the memory data register (mdr)
- ☐ On the software stack
- ☐ In the instruction register (ir)

✓ **Correct** 1/1 Points

8. What is the primary action of the 'ret' microcode sequence? 

- ☐ It pops the program counter off the stack.
- ☒ It restores the program counter (PC) with the value in register 15.
- ☐ It branches to the address stored in the branchTarget register.
- ☐ It increments the program counter by 4.

✓ **Correct** 1/1 Points

9. When executing a 'beq' (branch if equal) instruction, which hardware flag does the microcode test to determine if the branch should be taken? 

- ☐ `flags.GT`
- ☐ `flags.Z`

☒ flags.E

☐ flags.C

✓ **Correct** 1/1 Points

10. During the execution of a load (ld) or store (st) instruction, what is the ALU explicitly used for in the microcode? 

☐ To transfer data from the memory data register (mdr) to the register file.

☒ To calculate the effective memory address.

☐ To increment the program counter.

☐ To format the data before storing it in memory.



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